



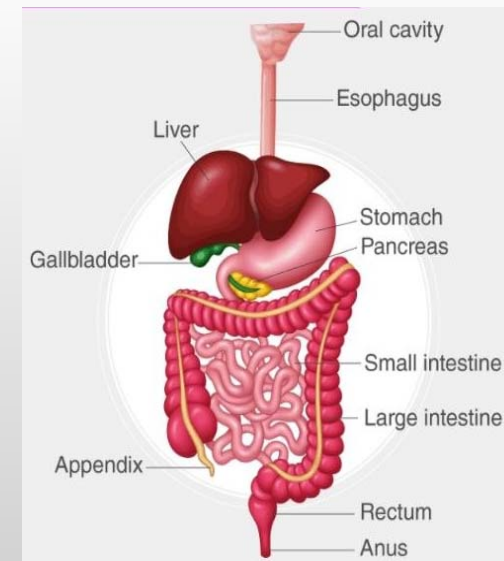
Systemic pathology Pathology of the GIT

Lec. 13

Dr.Zainab Hashim

Gastrointestinal tract (GIT)

- The gastrointestinal tract consists of the oral cavity , oropharynx, esophagus, stomach, small intestine, colon, rectum, and anus. Each region has unique, complementary, and highly integrated functions that together serve to regulate the intake, processing, and absorption of ingested nutrients and the disposal of waste products.
- Many pathological processes can affect the constituents of the gastrointestinal tract.

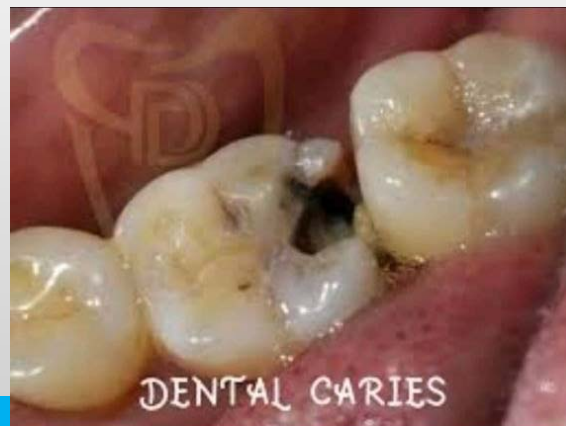


Oral Cavity

- Pathologic conditions of the oral cavity can be broadly divided into diseases affecting teeth their support structures, oral mucosa, salivary glands, and jaws.

1.Diseases of teeth and supporting structures:

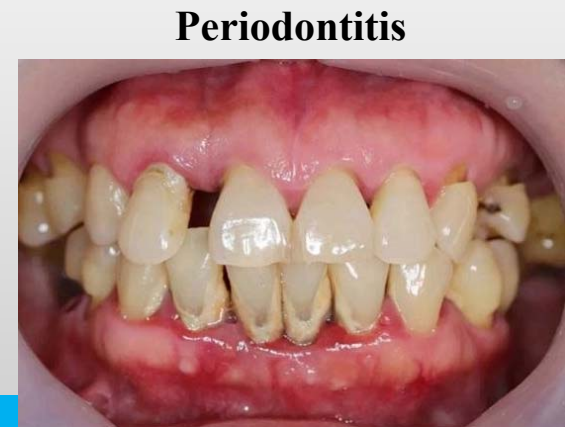
Dental caries : results from focal demineralization of tooth structure (enamel and dentin) caused by acids generated during the fermentation of sugars by bacteria.



Gingivitis : is a common and reversible inflammation of the mucosa surrounding the teeth. It is associated with buildup of dental plaque and calculus.



Periodontitis: is an inflammatory process that affects the supporting structures of the teeth periodontal ligaments, alveolar bone, and cementum. With progression, periodontitis may result in destruction of periodontal ligament and alveolar bone and eventual tooth loss.



2. Oral inflammatory lesions:

Aphthous Ulcers: are painful superficial ulcers of unknown etiology that may in some cases be associated with systemic diseases. In most cases they resolve spontaneously in 7 to 10 days but can recur.

Herpes simplex virus infections: the herpes simplex virus causes a self-limited infection that presents with vesicles that rupture and heal, without scarring.

Aphthous Ulcers



Herpes simplex infections



Fibromas



3. Proliferative lesions :

Fibromas : are submucosal nodular fibrous tissue masses that are formed when chronic irritation results in reactive connective tissue hyperplasia .They occur most often on the buccal mucosa along the bite line.

Pyogenic granuloma :is common reactive lesions of the oral mucosa typically found on the gingiva of children, young adults, and pregnant women .These lesions are richly vascular and typically ulcerated, which gives them a red to purple colour.

Pyogenic granuloma



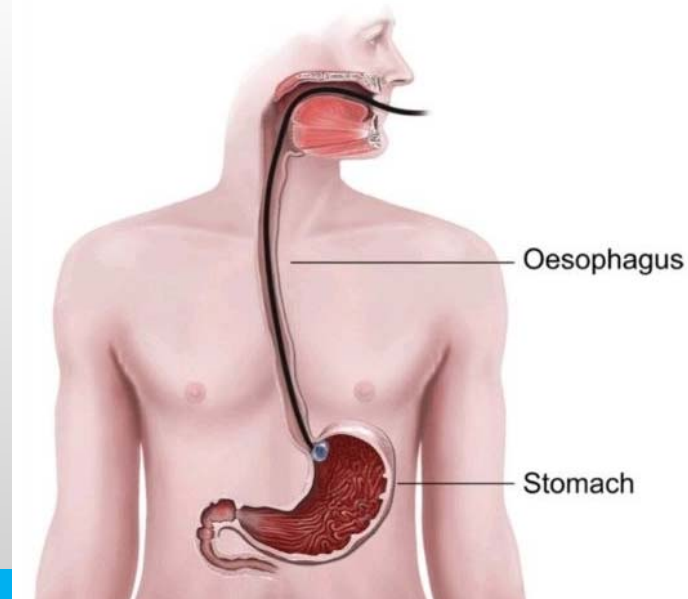
4. Neoplastic lesions:

Squamous cell carcinoma: an aggressive epithelial malignancy, is the sixth most common neoplasm in the world today and may arise anywhere in the oral cavity. However, the most common locations are the ventral surface of the tongue, floor of the mouth, lower lip, soft palate, and gingiva . It is classically linked to tobacco and alcohol use.



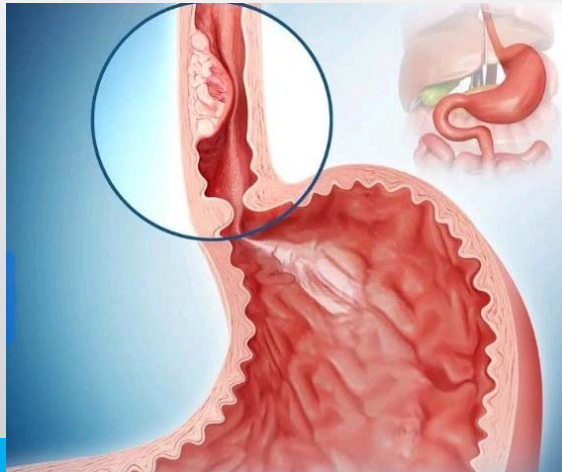
Esophagus

- It is a hollow, highly distensible muscular tube that extends from the epiglottis to the gastroesophageal junction. The main functions of the esophagus are to:
 1. Conduct food and fluids from the pharynx to the stomach
 2. Prevent reflux of gastric contents into the esophagus.



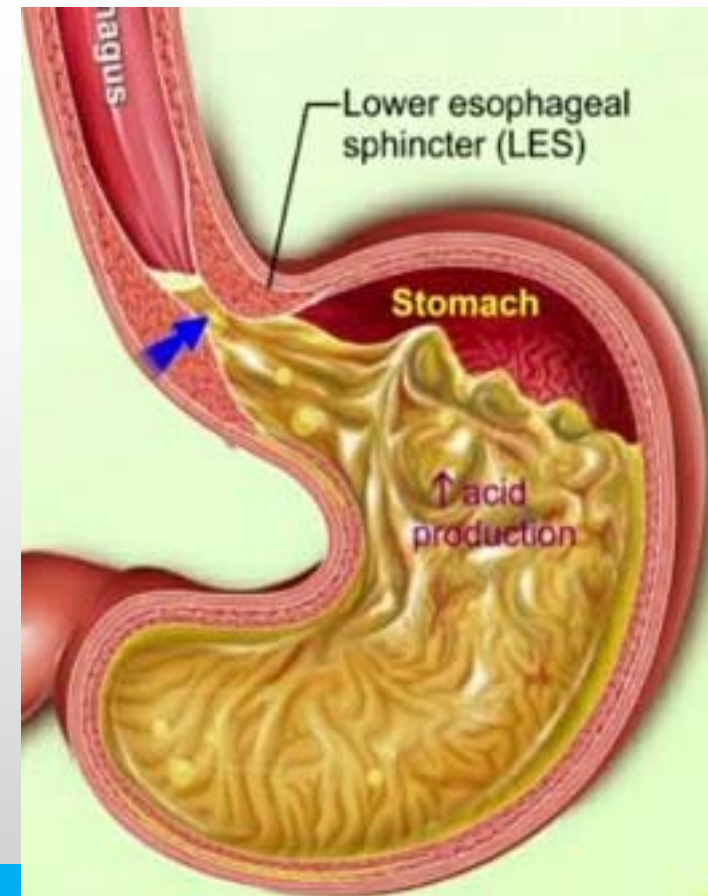
Examples of esophageal diseases

- **Esophageal stenosis** : consists of fibrous thickening of the esophageal wall. may be congenital or more commonly acquired. It is more frequently the result of severe esophageal injury with inflammatory scarring, as from gastroesophageal reflux disease (GERD) and caustic injury. Stenosis usually manifests as progressive dysphagia , at first to solid food but eventually to fluid as well.



Reflux Esophagitis (Gastroesophageal Reflux Disease or GERD):

- Is the most frequent cause of esophagitis due to reflux of gastric contents into the lower esophagus . The action of gastric juices is vital to the development of esophageal mucosal injury , because the stratified squamous epithelium of the esophagus is resistant to abrasion from foods but is sensitive to acid.



- Conditions that decrease lower esophageal sphincter tone or increase abdominal pressure contribute to GERD and include alcohol and tobacco use, obesity and pregnancy.

